Ex-No:8
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NMAP To Discover live hosts using NMAP Scans carry ICMP, TCP(UDP) ON TRY hack me platform

Aim:-

To study the working of the Nmap tool and discover live hosts in a networks using different host discovery techniques such as ARP scanning, ICMP Scanning, TCP Scanning and UDP Scanning on the try hack me platform. The experiment also aims to analyse the behaviour of different protocols and compare the result obtained from various scanning methods.

Steps:-

1) open the Try Hackme platform and select the Nmap room.

2) study the network topology and identify the source and destination hosts.

3) use ARP requests to discover the MAC address of devices available in the local Network

4) observe how ARP packets are brodcast with in the network and identify the devices the receive the request.

Output:

ARP:

sudo nmap -PR -sn 10.10.210.0/24

starting Nmap

Nmap scan report from 10.10.210.1

Host is up

Nmap done

ICMP:

nmap -PE -sn 10.10.210.0/24

starting Nmap

Nmap scan report for 10.10.210.6

Host is up

Nmap done:

TCP:

nmap -PS22-23 -sn 10.10.210.0/24

sudo nmap -PA80 -sn 10.10.210/24

5) use the ICMP protocol to send ping requests between two system and observe the responses.

6) Analyses the packet flow and identify the exchange of ARP packets before ICMP communication.

7) perform ARP host discovery using the Nmap command and identify all active hosts

8) perform ICMP Echo scanning and observe the hosts that respond to the scan.

9) Perform ICMP timestamp scanning and analyse the response generated by the target system.

10) Perform ICMP Address mask scanning and compare the result with the previous scans.

11) Perform TCP SYN scanning to identify live hosts by analyzing SYN and ACK packets.

12) Perform TCP ACK scanning to verify the accessibility of the target hosts.

13) Perform UDP scanning and observe how Nmap identifies active hosts through UDP packets

14) Perform a Reverse DNS lookup to obtain the host names of the discovered systems.

15) compare the result obtained from ARP, ICMP, TCP and UDP scanning techniques.

16) Analyse the advantages and limitations of each scanning method.

Result:-

Thus, live hosts in the network were successfully identified using different Nmap scanning techniques, ADP scanning was found to be effective for discovering devices in the local network while ICMP, TCP and UDP scanning techniques were useful for identifying active hosts and verifying network connectivity. the experiment also demonstrated the effectiveness of the Nmap tool in network discovery and analysis.